Satyabrata Das

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Summary

MS AI student at the University of Florida specializing in computer vision, deep learning, and real-time AI systems. Experienced in building scalable ML pipelines, GPU-accelerated inference systems, and production-grade AI applications using PyTorch, OpenCV, FastAPI, and Docker.

Education

University of Florida — **Aug 2025 - May 2027**
M.S., Artificial Intelligence Systems — *Gainesville, FL*
- **Coursework:** AI Safety & Robustness, AI Systems, Machine Learning for AI, Image Processing & Computer Vision, Applied Deep Learning

Experience

Movement Estimation & Analysis Lab (MEA Lab), University of Florida — **Feb 2026 - Present**
Machine Learning Research Assistant
- Developed computer-vision pipelines using MediaPipe, OpenCV, and Python to automatically analyze movement in over 100 clinical videos, reducing processing time and enabling consistent data extraction
- Engineered biomechanical feature-extraction systems-including joint-angle, symmetry, and temporal motion analytics-using Python and CUDA, which supplied inputs to ML models and improved severity-prediction accuracy
- Trained and evaluated PyTorch regression models for automated movement severity estimation, improving scoring consistency while reducing manual analysis effort.

ARC Document Solutions — **Jan 2022 - Sep 2024**
Software Engineer — *India*
- Built scalable real-time applications with Swift, WebSockets, and REST APIs for production AI platforms, enabling faster data processing and improving user response times.
- Developed automation and backend-integrated workflows supporting scalable application performance and reliability.

SCI-BI Software Pvt Ltd — **Jul 2021 - Dec 2021**
Software Trainee — *India*
- Developed ETL and automated data-processing pipelines handling millions of records daily with 99%+ processing accuracy.
- Optimized SQL-based data workflows and monitoring systems using Bash scripts and Prometheus/Grafana, improving large-scale data availability and processing efficiency.

Skills

- **AI & ML**: PyTorch, TensorFlow Lite, HuggingFace Transformers, Scikit-learn, LLMs, NLP, CUDA
- **Computer Vision**: OpenCV, MediaPipe, YOLO, 3D Pose Estimation, Video Analytics
- **Generative AI & Retrieval**: RAG, Vector Databases, Embedding Models, Retrieval Systems
- **Programming**: Python, C++, Swift, SQL
- **Systems & Tools**: FastAPI, Docker, REST APIs, CI/CD, Git, Linux, Prometheus/Grafana, Weights & Biases

Projects

PoseInsight: Real-Time Movement Analysis System | GitHub
- Built real-time pose estimation pipelines for exercise classification from webcam and uploaded video streams using OpenCV and MediaPipe.
- Engineered temporal biomechanical features and trained LSTM-based movement quality classification models for automated form assessment.
- Optimized inference latency through asynchronous frame processing and containerized deployment workflows for real-time execution.

JuggleIQ: CV-Based Soccer Performance Analytics | GitHub
- Architected GPU-accelerated computer-vision pipelines generating 1,000+ frame-level inferences per session.
- Integrated YOLO object detection, pose estimation, and Kalman filtering for structured movement analytics.
- Built FastAPI backend services and Dockerized deployment workflows for scalable AI analytics delivery.

SummarIQ: Math-Aware Research Paper Summarizer | GitHub
- Developed transformer-based NLP pipelines using HuggingFace T5/BART models for section-wise scientific document summarization and equation-aware text generation.
- Built custom LaTeX extraction and preprocessing workflows handling 200+ academic papers with structured figure, table, and equation parsing.
- Optimized inference throughput using GPU acceleration, batch processing, and FastAPI deployment achieving sub-1.4s latency.

Achievements

- **NASA EMERGE Data Hackathon:**Winner

Certifications

- **Fundamentals of Deep Learning: NVIDIA**
- **Building Transformer-Based NLP Applications: NVIDIA**
- **Supervised Machine Learning: Regression and Classification: DeepLearning.AI**